SMART INDIA HACKATHON 2026

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Problem Statement ID – [ADD YOUR PS ID]

Problem Statement Title-

Small farmers guess at sowing, spraying, and selling decisions — no affordable, real-time, local-language advisor exists.

Literacy barriers and delayed expert access force farmers into costly guesswork. What can them help is a voice-enabled, sensor-driven AI companion for instant, localized guidance on crop health, weather, and markets

Theme- Agriculture, FoodTech & Rural Development

PS Category- Software/Hardware

Team ID- [ADD TEAM ID]

Team Name (Registered on portal) – NOVA_CORE

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KRISHI MITRA

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Proposed Solution

01
Voice + Camera Farm App

02
ESP32 Field Hardware Sensors

03
Market Price Trend Analysis

04
Seasonal Weather Intelligence

05
Contamination & Nowcast Alerts

Innovation & Uniqueness

- Season-long farm companion — plans, grows, protects, and sells with the farmer, not just a single-query chatbot.
- First fusion of on-field sensors with government pollution data to flag environmental risk, not just crop health.
- Voice + camera remove the literacy barrier; live sensors and government data replace guesswork and delayed experts.
- Turns generic daily weather into minutes-matter, location-specific action alerts for storms and contamination.

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TECHNICAL APPROACH

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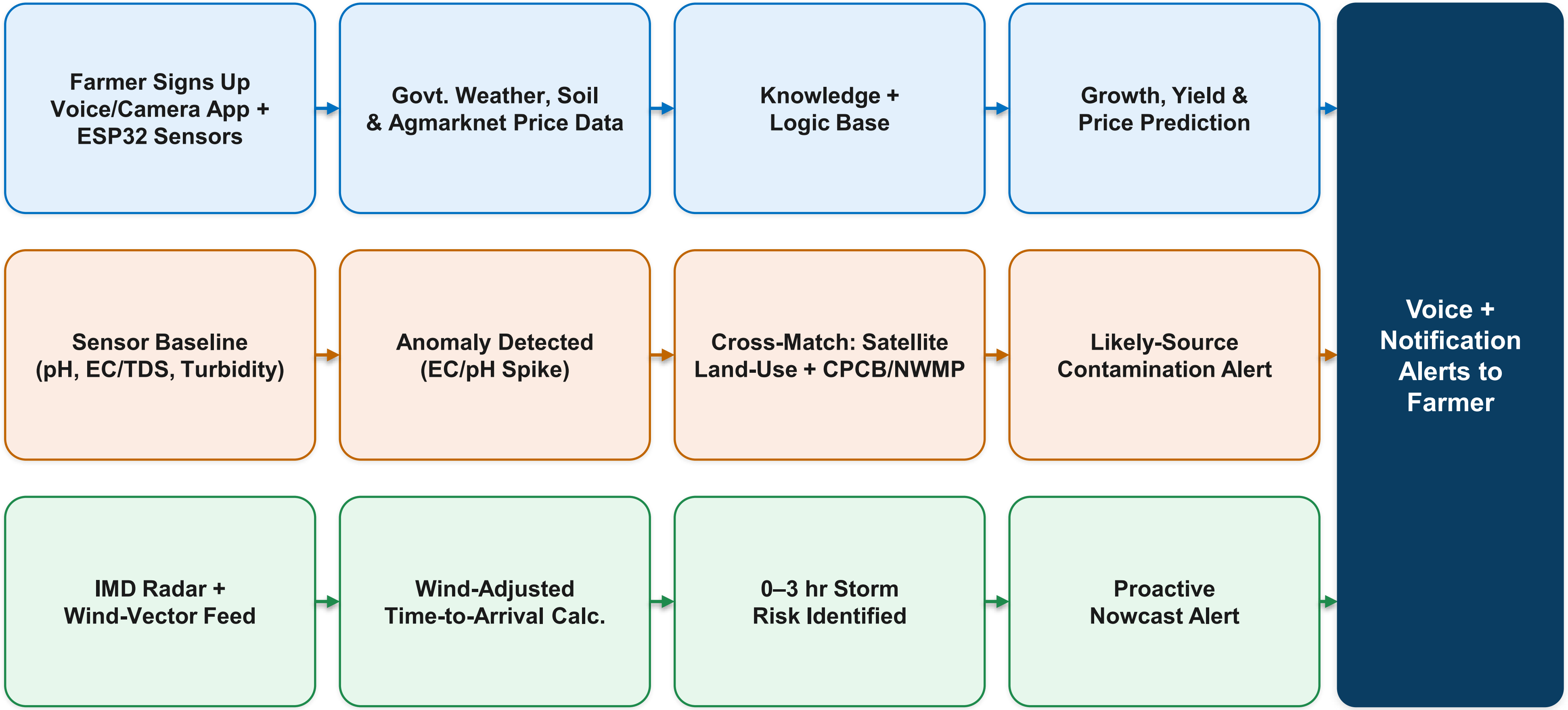
Technologies Used

- Voice + Camera App with ML Engine
- ESP32: Soil Moisture, NPK, DHT22
- Govt. Weather & Soil-Type APIs
- IMD Nowcast/Radar + Wind-Vector
- CPCB / NWMP Water Quality Data
- ISRO Bhuvan / Sentinel Land-Use
- Agmarknet / e-NAM Price APIs
- New Sensors: pH, EC/TDS, Turbidity, MQ-Gas

Methodology — Sensing to Alert Pipeline



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FEASIBILITY AND VIABILITY

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Feasibility

- All sensors (ESP32, NPK, moisture, DHT22, pH, EC/TDS, turbidity, MQ-gas) are low-cost, well-documented, and ESP32-compatible
- Govt. weather, soil, IMD nowcast & CPCB/NWMP data are public and already used on other platforms — no new infra needed
- Agmarknet / e-NAM mandi price data is public and updated regularly
- Realistically buildable within the hackathon timeline

Challenges & Risks

- Inconsistent field-sensor accuracy; cheap sensors flag anomalies but can’t confirm exact pollutants
- Patchy rural internet connectivity
- Gaps in govt. weather/soil data coverage across regions
- Nowcast/radar accuracy and pollution-data timeliness vary by region

Mitigation Strategy

- Calibrate sensors against manual soil/water tests
- Cache recent weather, soil & price data on-device; fall back to regional historical averages when live feeds are delayed
- Frame contamination alerts as “flag for farmer verification,” not a confirmed diagnosis
- Use wind-adjusted time-to-arrival estimates with a stated margin of error, not exact-minute promises

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IMPACT AND BENEFITS

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Farmers (Users)

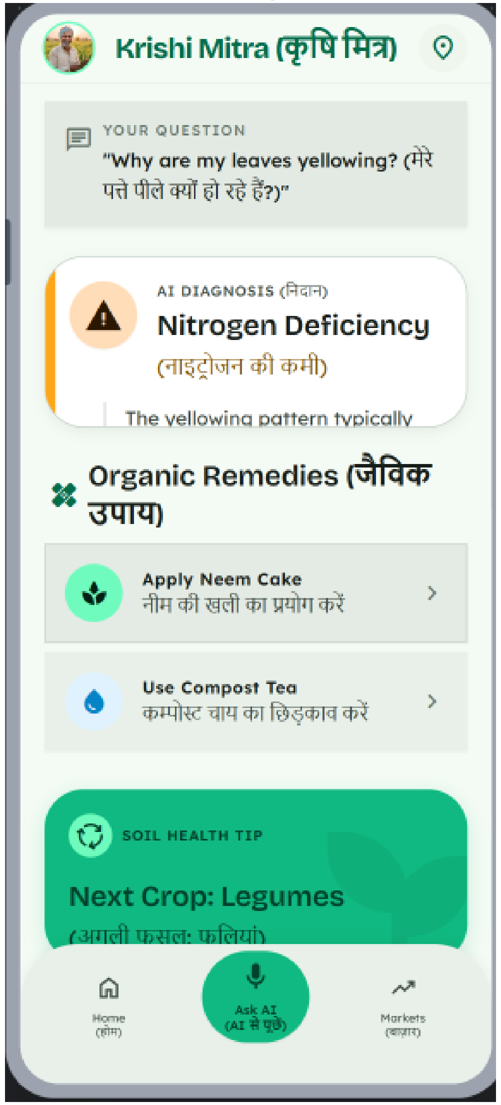
- **Direct, real-time access to expert-level guidance** on pesticides, fertilizers, crop cycles and growth periods.
- **Reduces dependence on guesswork** and inconsistent local advice.

Economic Impact

- **Higher yields and better-timed harvest/price decisions** raise farmer income.

Environmental & Social Impact

- **Data-driven fertilizer and pesticide use** protects soil quality.
- **Voice- and camera-based interaction** makes the assistant accessible even to farmers with low literacy.



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RESEARCH AND REFERENCES

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Research & Precedent:

- **IoT-enabled soil moisture prediction:** M. Afzal et al., “IoT-enabled adaptive watering system with ARIMA-based soil moisture prediction,” IEEE Access, vol. 13, 2025.
- **ML-based rice yield prediction in India:** D. De Clercq and A. Mahdi, “Feasibility of ML-based rice yield prediction in India at district level,” arXiv:2403.07967, 2024.
- **IoT soil sensing automation:** A. A. Abdelmoneim et al., “Internet of Things (IoT) for Soil Moisture Tensiometer Automation,” Micromachines, 14(2), 2023.
- **Voice-based AI farm advisory at scale:** N. Singh et al. (Digital Green & Microsoft Research), “Farmer.Chat: Scaling AI-Powered Agricultural Services,” arXiv:2409.08916, 2024.
- **AI advisory deployment learnings (incl. Bihar, India):** S. Collis et al. (Gates Foundation / GIZ / CLEAR Global), “Building AI-based advisory services for smallholder farmers,” arXiv:2601.11537, 2026.

Web References (for citation):

- **data.gov.in** – agricultural, weather & soil datasets.
- **mausam.imd.gov.in** – real-time weather API.
- **icar.org.in** – soil-type & crop-suitability data.
- **espressif.com** – ESP32/NPK/DHT22 hardware refs.

Real-World Case Studies:

- **Farmer.Chat (Digital Green)** – voice/text/image AI advisory chatbot for smallholder farmers; India, Kenya, Ethiopia & Nigeria.
- **Bharat-VISTAAR (Govt. of India)** – official voice-first AI farming platform; AgriStack + ICAR + weather data, launched 2026.
- **KissanAI** – voice-based AI copilot for farmers in their native language; 100,000+ users within months.